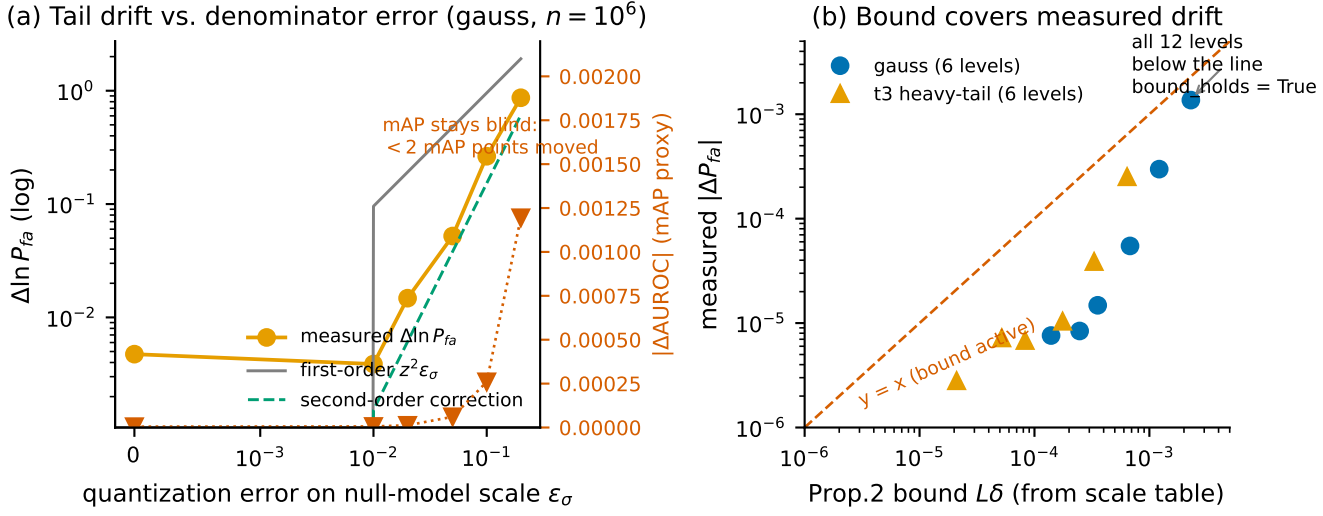
(a) Tail drift vs. denominator error (gauss, $n = 10^6$)
$\Delta \ln P_{fa}$ (log)
10^0
10^{-1}
10^{-2}
0.00200
0.00175
0.00150
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
|ΔAUROC| (mAP proxy)
mAP stays blind:
< 2 mAP points moved
measured $\Delta \ln P_{fa}$
first-order $z^2 \varepsilon_\sigma$
second-order correction
0
10^{-3}
10^{-2}
10^{-1}
quantization error on null-model scale ε_σ
(b) Bound covers measured drift
all 12 levels
below the line
bound holds = True
gauss (6 levels)
t3 heavy-tail (6 levels)
measured $|\Delta P_{fa}|$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
y = x (bound active)
10^{-6}
10^{-5}
10^{-4}
10^{-3}
Prop.2 bound $L\delta$ (from scale table)